
Prediction of Students' Dropout and Academic Success

Quantitative Analysis for Business Project

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Executive Summary

A comprehensive analysis of the "Predict Students Dropout and Academic Success" dataset was conducted, including descriptive statistics to examine patterns among the target variables: Dropout, Enrolled, and Graduate. Linear regression models and correlation analyses were implemented to understand the relationships between key variables. These methods provided valuable insights into the factors influencing student outcomes, enabling stakeholders and educational institutions to identify the underlying reasons for student dropouts and the critical factors that may help prevent them. The findings from this analysis offer actionable recommendations to assist institutions in understanding their customer segments and market behavior, ultimately supporting strategic interventions to improve retention and enhance academic success.

Introduction

Higher education institutions worldwide face significant challenges with student dropout and academic failure, which not only impact institutional performance but also affect students' personal and professional development. In response to this issue, the "Predict Students Dropout and Academic Success" dataset was created to enable the early identification of students at risk of dropping out or failing in their academic journey. By leveraging machine learning techniques, this dataset aims to provide actionable insights to address these challenges and implement timely support strategies.

The dataset, sourced from the UCI Machine Learning Repository, is designed for a three-category classification task:

1. Dropout – Students who discontinue their studies.
2. Enrolled – Students who remain active in their courses.
3. Graduate – Students who successfully complete their degrees within the normal duration.

Each instance in the dataset represents an individual student and includes information known at the time of enrollment, encompassing academic paths, demographics, and socio-economic factors. The absence of missing values ensures the dataset's completeness, making it suitable for robust analysis and modeling.

This report explores the dataset, evaluates its features, and applies machine learning methods to predict academic outcomes. The ultimate goal is to uncover patterns and build predictive models that can assist higher education institutions in identifying at-risk students and implementing tailored interventions to improve student success and reduce dropout rates.

The primary stakeholders of this project are higher education institutions, including university administrators, faculty, academic advisors, and policy makers. These groups will benefit significantly as the insights from this analysis can help identify students at risk of dropout, improve graduation rates, and enhance overall student success. Additionally, students themselves are direct beneficiaries, as early interventions can provide the necessary support to overcome challenges and ensure academic continuity.

Practical recommendations from this project include implementing early warning systems using machine learning models to flag at-risk students based on socio-economic and academic factors. Institutions should offer targeted support services such as tutoring, counseling, and financial aid while enhancing monitoring systems to track student progress regularly. Resource allocation strategies, including mentorship programs and policy adjustments, can address specific challenges and reduce dropout rates.

Key actions need to be taken by university administrators, who should deploy predictive systems and allocate resources for support programs. Faculty and academic advisors must engage flagged students with personalized guidance, while student affairs teams should roll out workshops and counseling services. Policy makers and educational leaders play a role in creating long-term strategies to support retention goals, ensuring that institutions foster an inclusive and supportive learning environment.

Methods and Data

Data characteristics

In order to comprehend data variables, descriptions can be seen in Figure 1.1:

Variable Selection	Type	Description
Marital Status	Integer	1 – single 2 – married 3 – widower 4 – divorced 5 – facto union 6 – legally separated
Daytime/evening attendance	Integer	1 – daytime 0 - evening
Nationality	Integer	1 - Portuguese; 2 - German; 6 - Spanish; 11 - Italian; 13 - Dutch; 14 - English; 17 - Lithuanian; 21 - Angolan; 22 - Cape Verdean; 24 - Guinean; 25 - Mozambican; 26 - Santomean; 32 - Turkish; 41 - Brazilian; 62 - Romanian; 100 - Moldova (Republic of); 101 - Mexican; 103 - Ukrainian; 105 - Russian; 108 - Cuban; 109 - Colombian
Admission grade	Continuous	Admission grade (between 0 and 200)
Debtor	Integer	1 – yes 0 – no
Tuition fees up to date	Integer	1 – yes 0 – no
Gender	Integer	1 – male 0 – female
Scholarship holder	Integer	1 – yes 0 – no
Age at enrollment	Integer	Age of student at enrollment
International	Integer	1 – yes 0 – no
Unemployment rate	Continuous	Unemployment rate (%)
Inflation rate	Continuous	Inflation rate (%)
Target	Categorical	Target. The problem is formulated as a three category classification task (dropout, enrolled, and graduate) at the end of the normal duration of the course

Figure 1.1

Correlation Matrix

To effectively identify and analyze potential variables for deeper exploration within the dataset, a correlation matrix which is Figure 1.2 has been constructed. This approach allows for the evaluation of relationships between variables, highlighting any strong correlations beyond those directly involving the target variable. By identifying these interdependencies, we can pinpoint key predictors or redundant features, thus facilitating a more focused and insightful analysis.

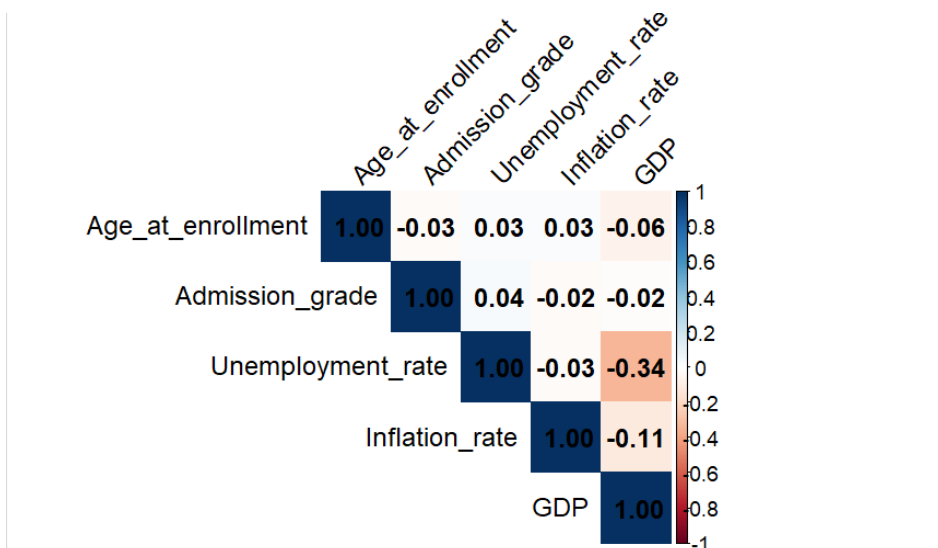


Figure 1.2

The Figure 1.2 provides a clear understanding of the relationships between the given variables: Age at Enrollment, Admission Grade, Unemployment Rate, Inflation Rate, and GDP. Correlation coefficients range between -1 and 1, where values close to 1 or -1 indicate strong positive or negative correlations, respectively, while values near 0 suggest weak or no correlation. The matrix reveals that most variables in this dataset exhibit low correlations with each other.

The strongest observed relationship in the matrix is a moderate negative correlation of -0.34 between the Unemployment Rate and GDP. This suggests that as unemployment decreases, GDP tends to increase, which aligns with economic expectations, as higher employment rates generally contribute positively to economic growth. This relationship warrants further exploration as it could hold significant predictive value in economic or policy-related analyses.

Other variables, such as Age at Enrollment and Admission Grade, show weak correlations with all other features, with values ranging between -0.03 and 0.04. This indicates a lack of significant linear relationships between these variables and the others in the dataset. While no strong correlations are evident outside the Unemployment Rate-GDP pair, it is essential to analyze these findings in the broader context of the data to determine their relevance and impact.

Descriptive Statistics

Variable	Vars	N	Mean	SD	Median	Trimmed	MAD	Min	Max
Course	4	4424	8856.64	2063.57	9238	9330.4	388.44	33	9991
Previous qualification (grade)	7	4424	132.61	13.19	133.1	132.31	10.53	95	190
Admission grade	13	4424	126.98	14.48	126.1	126.31	12.45	95	190
Age at enrollment	20	4424	23.27	7.59	20	21.57	2.97	17	70
Unemployment rate	34	4424	11.57	2.66	11.1	11.5	2.52	7.6	16.2
Inflation rate	35	4424	1.23	1.38	1.4	1.18	1.78	-0.8	3.7
GDP	36	4424	0	2.27	0.32	0.09	2.18	-4.06	3.51

Figure 2.1

The Figure 2.1 represents descriptive statistics for various variables based on a sample size of 4424 observations. For the Course variable, the mean value is 8856.64 with a standard deviation (SD) of 2063.57, and a median of 9238. The values range widely from a minimum of 33 to a maximum of 9991, indicating a large variability. For Previous qualification (grade), the mean is 132.61, with a relatively small SD of 13.19, and the values range between 95 and 190, showing moderate dispersion around the mean.

The admission grade variable has a mean of 126.98 and an SD of 14.48, with a median of 126.1, closely aligning with the trimmed mean of 126.31. The range is 95 to 190, showing consistency in student grades. The Age at enrollment has a mean age of 23.27 with a notable SD of 7.59, and

the median age is 20. The range spans from 17 to 70, suggesting a mix of younger and older enrollees, but the trimmed mean (21.57) indicates slightly reduced variability.

For macroeconomic indicators, the Unemployment rate has a mean of 11.57 and an SD of 2.66, with values ranging from 7.6 to 16.2, showing moderate dispersion. The Inflation rate has a lower mean of 1.23 and a higher relative SD of 1.38, with values spanning from -0.8 to 3.7, indicating occasional deflation. Lastly, the GDP variable has a mean of 0, an SD of 2.27, and values ranging between -4.06 and 3.51, reflecting a balance around zero but considerable variability. This table provides a comprehensive summary of central tendencies, dispersion, and ranges for each variable.

Marital Status

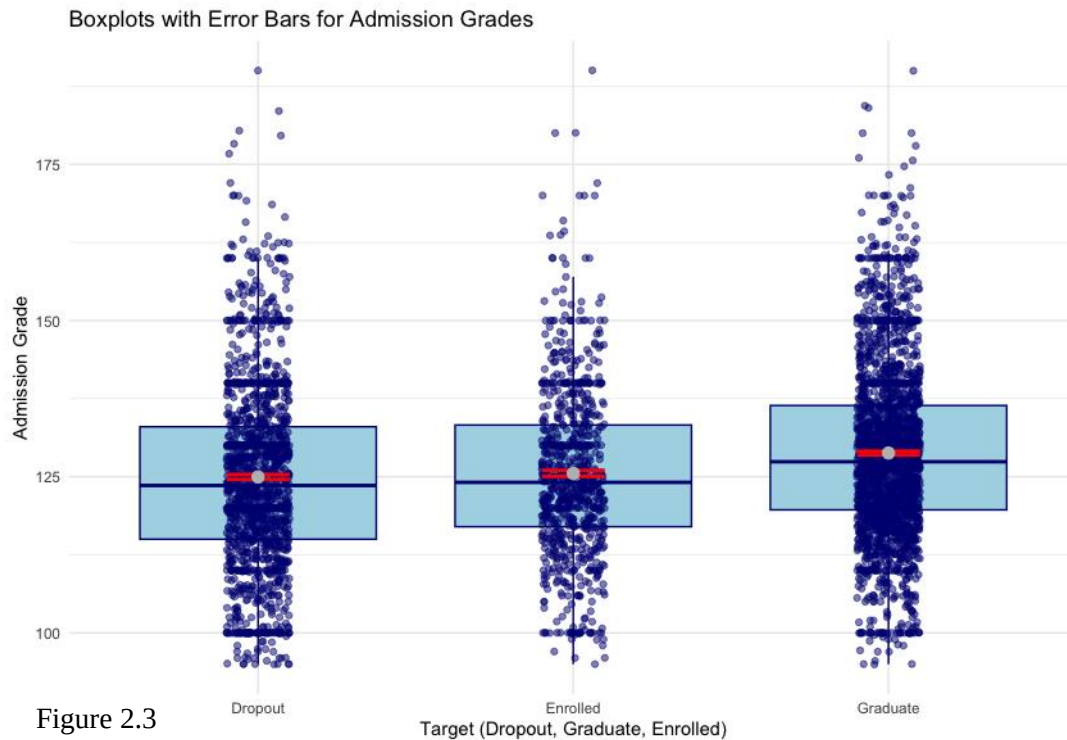
The marital status data presented shows the following distribution among students:

Marital Status					
Single	Married	Widower	Divorced	Facto Union	Legally Separated
3919	379	4	91	25	6

Figure 2.2

The Figure 2.2 shows the distribution of Marital Status across six categories, with Single individuals being the largest group at 3919, followed by Married at 379. The Divorced group includes 91 individuals, while those in Facto Union account for 25. The categories of Legally Separated and Widower are the smallest, with 6 and 4 individuals, respectively. Overall, the majority are single, with the remaining groups having significantly smaller representation.

Admission Grade



The plot displays (Figure 2.3) boxplots with error bars and jittered points for Admission Grades across three categories: Dropout, Enrolled, and Graduate. The central boxes show the interquartile range (IQR), with medians around 125 for all groups, while the red points and error bars indicate the mean Admission Grades with standard errors. The distributions are similar across categories, with most values clustered between 110 and 140, though some outliers exceed 150 or fall below 100. Overall, there is no visually significant difference in Admission Grades between the groups.

Correlation between Admission Grade and Target

The correlation coefficient of 0.039 between Admission Grade and Target indicates a very weak positive correlation. This suggests that there is almost no linear relationship between Admission Grades and the target categories (Dropout, Enrolled, Graduate). In practical terms, the admission grade does not appear to strongly influence educational outcomes.

The Figure 2.3 supports this observation, as the distributions of Admission Grades across the three target categories are visually very similar. All groups have medians around 125, with values clustering between 110 and 140. Additionally, there are no significant differences in the spread of the data, and all categories share similar patterns of outliers.

Daytime / Evening Attendance

Day Time Evening Attendance	
Daytime	Evening
483	3941

Figure 2.4.1

The data reveals a total of 3941 evening attendances and 483 daytime attendances across all target categories (Dropout, Enrolled, and Graduate) without any additional filters applied (Figure 2.4.1). To gain deeper insights into the patterns of attendance, further analysis is shown below, including comparisons between daytime and evening attendance based on target variables. By examining these patterns, we can identify trends or disparities that may exist between the two attendance types, providing valuable information to inform decision-making or policy development. A detailed breakdown or analysis will highlight any significant relationships between attendance times and key performance indicators.

Target	Daytime / Evening	Count	%
Dropout	Evening	207	4.68%
Dropout	Daytime	1214	27.44%
Enrolled	Evening	75	1.70%
Enrolled	Daytime	719	16.25%
Graduate	Evening	201	4.54%
Graduate	Daytime	2008	45.39%

Figure 2.4.2

The Figure 2.4.2 provides a breakdown of the target categories (Dropout, Enrolled, Graduate) by Daytime/Evening along with their respective counts and percentages. For Dropouts, the majority occurred during the Daytime (1214; 27.44%) compared to the Evening (207; 4.68%). Similarly, among those Enrolled, most are in the Daytime (719; 16.25%), while the Evening accounts for a much smaller percentage (75; 1.70%). For Graduates, the trend is also significant, with Daytime graduates dominating (2008; 45.39%) and Evening graduates being far fewer (201; 4.54%). This suggests that participation, regardless of target outcome, is predominantly concentrated in the Daytime, particularly among graduates, highlighting a significant preference or opportunity disparity between Daytime and Evening programs.

Debtor and Tuition Fees Up to Date

Target	Debtor	Tuition Fees Up to Date	Count
Dropout	No	No	242
Dropout	No	Yes	867
Dropout	Yes	No	215
Dropout	Yes	Yes	97
Enrolled	No	No	22
Enrolled	No	Yes	682
Enrolled	Yes	No	20
Enrolled	Yes	Yes	70
Graduate	No	No	18
Graduate	No	Yes	2090
Graduate	Yes	No	11
Graduate	Yes	Yes	90

Figure 2.5.1

The Figure 2.5.1 summarizes the distribution of Debtor status and Tuition Fees Up to Date across three target categories: Dropout, Enrolled, and Graduate. Among Dropouts, the majority (867) have their tuition fees up to date but are not debtors. For Enrolled students, most (682) are not debtors and have their fees up to date. Among Graduates, a significant majority (2090) have their fees up to date and are not debtors, while a small number (90) are debtors with fees up to date. Overall, having tuition fees up to date correlates with better outcomes, particularly for graduates, while debtors with unpaid fees are less represented among positive outcomes.

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.3425	0.1173	0.1169	<2.2e-16

Model	B	Std. Error	T	Sig.
(Intercept)	0.23241	0.03244	7.165	9.05E-13
Tuition_fees_up_to_date	0.71578	0.0331	21.624	<2e-16
Debtor	-0.03948	0.03381	-1.168	0.243

Figure 2.5.2

The regression model (Figure 2.5.2) explains 11.73% of the variance in the dependent variable, as indicated by the R^2 value. The predictor Tuition_fees_up_to_date has a strong positive effect ($B = 0.71578$, $p < 2e-16$), meaning that for every 1-unit increase in Tuition_fees_up_to_date, the dependent variable increases by approximately 0.716 units, making this predictor highly significant. On the other hand, the predictor Debtor has a small negative effect ($B = -0.03948$) but is not statistically significant ($p = 0.243$), suggesting that it does not

contribute meaningfully to the model. The intercept is 0.23241 and statistically significant ($p < 0.001$), representing the baseline value when all predictors are zero. Overall, the model has a moderate fit, with Tuition_fees_up_to_date being the key driver of the dependent variable.

Based on the updated data, the regression model's results align with observable trends. Tuition_fees_up_to_date plays a significant role, as seen by its strong positive impact on the target outcome. For instance, among graduates, 2090 individuals had their tuition fees up to date compared to only 18 graduates without up-to-date fees. This emphasizes that having fees up to date is a strong predictor of success (graduation). On the other hand, the Debtor status appears to have less influence, as both dropouts and enrolled students show mixed patterns across debtor categories.

Age at Enrollment

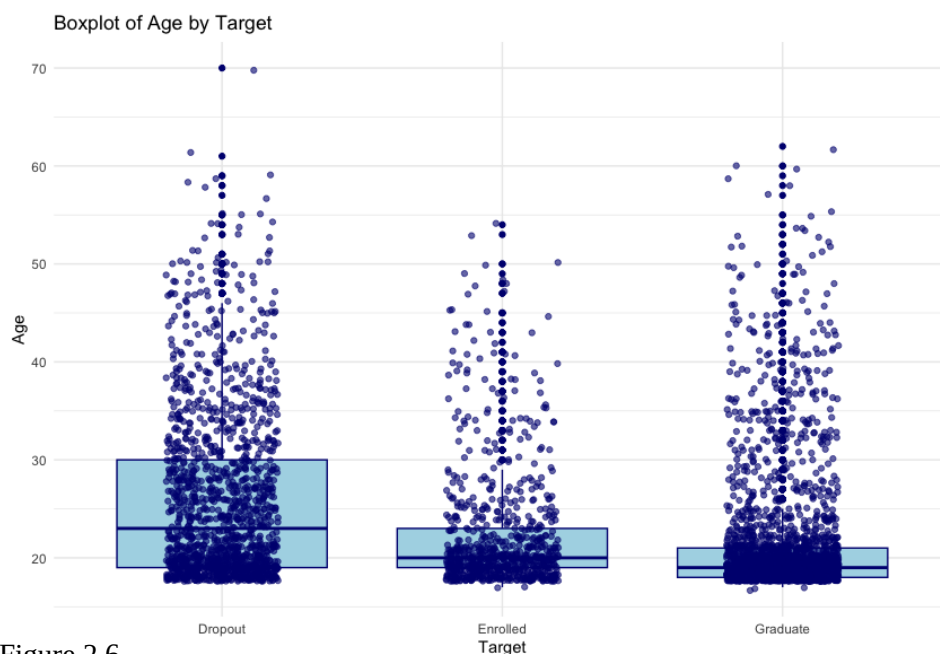


Figure 2.6

The Figure 2.6 reveals age-related differences across the target categories: Dropout, Enrolled, and Graduate. The dropout group has the highest median age and the widest age range, with a notable presence of older individuals, including outliers above 60 years. The enrolled group shows a younger and more concentrated age distribution, with a lower median age. In contrast, the graduate group has an intermediate pattern, with a slightly higher median age than the enrolled group and a wider spread, including some older graduates. Overall, the data suggests that age may play a role in educational outcomes, with older individuals more likely to drop out or graduate, while younger individuals tend to be enrolled.

Correlation between Age and Target

The correlation coefficient of -0.202 between age and target (educational outcomes) indicates a weak to moderate negative correlation. This suggests that as age increases, individuals are slightly more likely to fall into less favorable educational categories, such as Dropout, and less likely to be in the Enrolled group.

The Figure 2.6 supports this relationship: the Dropout group has the highest median age and the widest age range, including a significant number of older individuals, highlighting that older age is associated with dropping out. Conversely, the Enrolled group has a younger and more concentrated age distribution, suggesting that younger individuals are more likely to remain enrolled. The Graduate group has an intermediate pattern, with a slightly higher median age than the Enrolled group but still younger overall compared to the Dropout group.

Gender

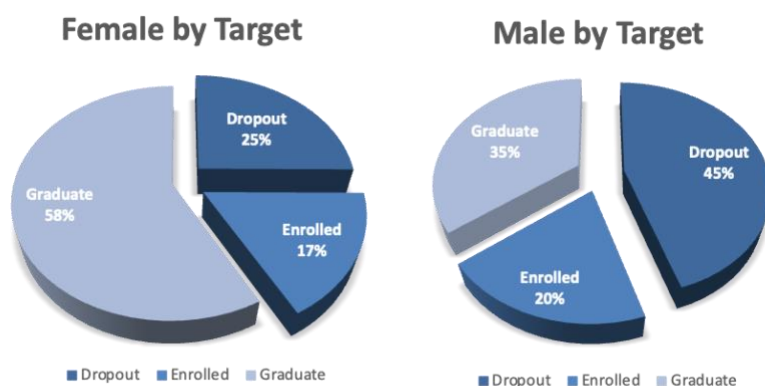


Figure 2.7

The Figure 2.7 compares the educational outcomes of females and males based on three categories: Dropout, Enrolled, and Graduate. For females, the majority (58%) have graduated, with 25% dropping out and 17% currently enrolled. In contrast, for males, only 35% have graduated, while a higher percentage (45%) have dropped out, and 20% are still enrolled. This highlights a significant gender disparity, with females achieving higher graduation rates and lower dropout rates compared to males. The data suggests that males are more likely to drop out and less likely to graduate than their female counterparts.

Correlation between Gender and Target

The correlation coefficient of -0.118 between gender and target (educational outcomes) indicates a weak negative correlation. This means that as the variable representing gender changes (e.g., from female to male), there is a slight tendency for educational outcomes to worsen, such as higher dropout rates and lower graduation rates.

The pie chart data supports this correlation: females have significantly better educational outcomes, with 58% graduating compared to only 35% of males. Conversely, males exhibit higher dropout rates (45%) compared to females (25%). The negative correlation, although weak, reflects this trend—males are more likely to drop out and less likely to graduate, while females are more likely to graduate and less likely to drop out.

In Depth Analysis and Result

This section will provide a detailed explanation of Regression Models and ANOVA, with a focus on analyzing the target variable as the dependent variable. Both basic and multiple logistic regression models will be implemented to examine the relationships and significance of various independent variables on the target variable. By leveraging these statistical methods, the analysis will aim to uncover patterns, determine the strength of associations, and assess the predictive power of key variables influencing the target outcomes. The results will provide valuable insights into the factors that significantly impact educational outcomes across the different target categories.

ANOVA					
Source	Sum of squares SS	df	Mean square MS	F	P-value
Model (Regression)	11.7	1	11.74	24.55	0.00000075
Error(Residual)	2114.4	4422	0.478		
Total	2126.1	4424	12.218		

Figure 3.1

The ANOVA result (Figure 3.1) indicates that the regression model is statistically significant, with an F-statistic of 24.55 and a p-value of 0.00000075, which is far below the conventional significance threshold of 0.05. This suggests that the model explains a portion of the variability in the response variable.

However, the Model Sum of Squares (11.7) is relatively small compared to the Total Sum of Squares (2126.1), indicating that while the regression model is significant, the amount of variance explained is minimal. Most of the variability remains unexplained, as reflected by the

large Error Sum of Squares (2114.4). Therefore, while the predictor in the model has a statistically significant effect, its practical contribution to explaining the outcome appears limited.

Model Summary				
Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	0.0743	0.00552	0.005297	0.6915

a. Predictors: Marital Status

Coefficients					
Model	Unstandardized Coefficients		Standardized Beta	T	Sig.
	B	Std. Error			
(Constant)	0.95851	0.02274	-	42.142	<2e-16
Marital_status	-0.08505	0.01716	-0.0743098	-4.955	7.50E-07

a. Dependent Variable: Target

Figure 3.2

The regression analysis (Figure 3.2) shows that Marital Status has a statistically significant negative relationship with the target variable. The unstandardized coefficient ($B = -0.08505$) indicates that for each one-unit increase in marital status, the target value decreases by 0.08505 units. The standardized coefficient ($Beta = -0.0743$) further confirms the small effect size. The t-statistic (-4.955) and corresponding p-value (7.50E-07) suggest that this relationship is highly significant, as the p-value is far below the conventional threshold of 0.05.

The overall model fit reveals a very weak relationship between marital status and the target variable. The R Square value (0.00552) indicates that marital status explains only 0.552% of the variance in the target variable, which is minimal. The adjusted R Square value (0.005297) confirms this finding after accounting for the model's degrees of freedom. Additionally, the R value (0.0743), which represents the correlation between the predictor and the outcome, is low, further emphasizing the weak predictive power of marital status in this context.

Despite the weak relationship, the model is statistically significant, as indicated by the F-statistic (24.55) and its associated p-value (7.497e-07). This suggests that marital status does have a measurable effect on the target variable, though its practical impact is negligible. The small R Square value implies that other variables not included in the model likely explain the majority of the variance. Therefore, while marital status contributes to the prediction, its effect is minimal, and additional predictors would be needed to improve the model's explanatory power.

Complete Model

Model Summary				
Model	R	R Square	Adjusted R Square	P-value
LM	0.3789	0.1436	0.1423	<2.2e-16

Model	Estimate	Std. Error	T	Pr(> t)
(Intercept)	0.5149956	0.1077406	4.78	1.81E-06
Age_at_enrollment	-0.0124466	0.0014882	-8.363	< 2e-16
Admission_grade	0.0008651	0.000668	1.295	0.1953
Gender	-0.0874146	0.0207537	-4.212	2.58E-05
Daytime_evening_atten	-0.0277783	0.0350588	-0.792	0.4282
Fees_up_to_date	0.6435706	0.0332613	19.349	< 2e-16
Debtor	-0.0269173	0.0333656	-0.807	0.4199
Scholarship_holder	0.0595135	0.0231166	2.57E+00	0.0101

Figure 3.3

Linear Equation of Complete Data (Figure 3.3)

$$Y = \beta_0 + \beta_1 (\text{Age at Enrollment}) + \beta_2 (\text{Admission Grade}) + \beta_3 (\text{Gender}) + \beta_4 (\text{Daytime/evening attendance}) + \beta_5 (\text{Fees up to date}) + \beta_6 (\text{Debtor}) + \beta_7 (\text{Scholarship holder}) + \epsilon$$

Linear Equation of Complete Calculated Data (Figure 3.3)

$$Y = 0.5149956 + (-0.012444) x (\text{Age at Enrollment}) + 0.0085651 x (\text{Admission Grade}) + (-0.087414) x (\text{Gender}) + (-0.027778) x (\text{Daytime/Evening Attendance}) + (-0.643570) x (\text{Fees Up to Date}) + (-0.0269173) x (\text{Debtor}) + 0.0595135 x (\text{Scholarship Holder}) + \epsilon$$

The linear regression model (Figure 3.3) summary indicates that the overall model is statistically significant with a very low p-value (<2.2e-16). However, the explanatory power of the model is modest, with an R-squared value of 0.1436, meaning that only about 14.36% of the variation in the dependent variable can be explained by the independent variables. This suggests that while the model identifies some relationships, it does not account for much of the variability in the data.

Several predictors in the model have statistically significant relationships with the dependent variable. Notably, Age_at_enrollment and Fees_up_to_date show strong negative associations, with increasing age at enrollment and having fees up to date being linked to lower values of the dependent variable. Scholarship_holder is also statistically significant and

positively related to the outcome, suggesting that holding a scholarship is associated with a higher value of the dependent variable. Gender is significant as well, with a negative coefficient, implying that gender might influence the outcome, though the exact nature of the relationship would depend on the coding of this variable.

On the other hand, some predictors do not appear to have meaningful effects. Admission_grade and Daytime_evening_attendance are not statistically significant, suggesting that factors like the grade at admission and whether a person attends during the day or evening do not substantially influence the dependent variable. Similarly, Debtor has a high p-value, meaning being a debtor does not significantly affect the outcome. These findings highlight that not all factors in the model are relevant, and further refinement or inclusion of additional variables might improve the model's explanatory power.

Variance Inflation Factor (VIF)

Variable	VIF	Interpretation
Age_at_enrollment	1.367892	This is a relatively low VIF, indicating that Age_at_enrollment is not highly correlated with other predictors, and multicollinearity is not a major concern for this variable.
Admission_grade	1.003882	This VIF is very close to 1, suggesting that Admission_grade has little to no correlation with other predictors in the model. This is ideal, as it indicates minimal multicollinearity.
Gender	1.053735	This also has a low VIF, indicating that Gender is not highly correlated with other variables in the model.
Daytime_evening_attendance	1.282625	This VIF suggests that Daytime_evening_attendance has a small correlation with other predictors but not enough to cause major concerns about multicollinearity.
Fees_up_to_date	1.247627	Similarly, this VIF indicates a small amount of multicollinearity with other predictors, but it's not problematic.
Debtor	1.203693	This value suggests that Debtor is also not strongly correlated with the other variables in the model.
Scholarship_holder	1.070506	This is another low VIF, indicating minimal multicollinearity with other predictors.

Figure 3.4

All the VIF values (Figure 3.4) provided are well below 5 (and most are closer to 1), which means there is little to no problematic multicollinearity among the predictors in the model. This suggests that the model's coefficient estimates are relatively stable and reliable. Generally, a VIF above 10 is considered high and would warrant further investigation, but that is not the case here.

Reduced Model

Model Summary				
Model	R	R Square	Adjusted R Square	P-value
LM	0.3782	0.143	0.1423	<2.2e-16

Model	Estimate	Std. Error	T	Pr(> t)
(Intercept)	0.57407	0.04582	12.528	< 2e-16
Age_at_enrollment	-0.01197	0.00132	-9.066	< 2e-16
Gender	-0.08809	0.02071	-4.254	2.15E-05
Fees_up_to_date	0.65727	0.0305	21.548	< 2e-16
Scholarship_holder	0.05957	0.02311	2.578	0.00998

Figure 3.5

The Model Summary (Figure 3.5) table provides key metrics about the linear regression model. The R-value of 0.3782 suggests a moderate positive correlation between the independent variables and the dependent variable, Target_Numeric. The R-Square value of 0.143 indicates that approximately 14.3% of the variance in the dependent variable is explained by the predictors included in the model. The adjusted R-Square value of 0.1423, which adjusts for the number of predictors, remains very close to the R-Square value, suggesting that the model does not overfit the data. The overall model significance, indicated by a P-value of < 2.2e-16, shows that the regression model as a whole is statistically significant.

The coefficients table (Figure 3.5) provides insight into the impact of each predictor on the dependent variable, Target_Numeric. The intercept of 0.57407 represents the baseline value of the dependent variable when all predictors are zero. The variable Age_at_enrollment has a coefficient of -0.01197 with a highly significant p-value (< 2e-16), indicating that for every additional year of age, the dependent variable decreases by approximately 0.012 units. Similarly, Gender has a negative coefficient of -0.08809 (p = 2.15e-05), meaning that being in a specific gender group lowers the dependent variable by approximately 0.088 units, holding other factors constant.

On the other hand, Fees_up_to_date has the largest positive coefficient of 0.65727 and is highly significant (p < 2e-16), showing that individuals with fees up to date have a substantial positive impact on the dependent variable. Scholarship_holder has a smaller yet statistically significant positive coefficient of 0.05957 (p = 0.00998), suggesting that holding a scholarship is associated with an increase of approximately 0.06 units in the dependent variable. Overall, the predictors Fees_up_to_date and Age_at_enrollment appear to have the strongest associations with the target variable, both statistically and practically.

Conclusion

The analysis conducted on the dataset "Predict Students' Dropout and Academic Success" revealed critical insights into the factors influencing student outcomes, namely dropouts, enrollments, and graduations. Descriptive statistics demonstrated that variables such as age at enrollment, tuition fee status, and gender exhibited measurable relationships with academic success. Specifically, having tuition fees up to date emerged as a significant predictor of graduation, with graduates being the largest group with current fees. Conversely, age displayed a weak negative correlation, suggesting that older students are more likely to drop out or face academic challenges. While other factors, such as marital status and attendance type, showed some significance, their predictive power remained relatively low.

The regression and ANOVA analyses further confirmed the moderate explanatory power of the model, with an R-squared value of approximately 14%. The predictors Tuition_fees_up_to_date and Scholarship_holder were found to have the most substantial impacts on positive academic outcomes, highlighting the importance of financial stability and institutional support. However, factors like Admission_grade and Debtor status were statistically insignificant, underscoring the need for institutions to focus on actionable predictors that truly influence retention and graduation rates. Gender disparities were also noted, with females achieving higher graduation rates and males exhibiting elevated dropout tendencies, reflecting broader social and academic trends.

In conclusion, the analysis provides actionable recommendations for educational institutions to enhance student success rates. Implementing early intervention systems based on tuition fee monitoring and scholarship programs can significantly improve retention. Additionally, targeted support for older students and gender-specific strategies may address the underlying challenges identified. While the current model explains only a portion of the variability, refining predictors and incorporating additional factors could enhance predictive accuracy and provide a more holistic approach to tackling student dropouts.

Appendix

Appendix A: Transformation of Categorical Target Variable

The original dataset contained a categorical target variable with three levels: Dropout, Graduate, and Enrolled. Since regression analysis requires numerical input, a new variable called `target_numeric` was created by mapping the categories to numerical values. Specifically, "Dropout" was coded as 0, "Graduate" as 1, and "Enrolled" as 2. This transformation enabled the use of regression techniques while preserving the categorical distinctions of the target variable.

Appendix B: Exclusion of Certain Variables

During the preliminary analysis, the variables GDP, Unemployment Rate, Inflation Rate, and Previous Qualifications were evaluated for their relationship with the target variable. The analysis revealed no significant correlation or meaningful statistical association between these variables and the target. As a result, they were excluded from further research to ensure the analysis focused on variables with stronger relationships or predictive potential.

Appendix C: Correlation Matrix Analysis

The correlation matrix (Figure 1.2) was utilized to identify significant variables for further analysis. However, the results obtained from the matrix did not provide insights directly related to our primary business question.

Citations

<https://archive.ics.uci.edu/dataset/697/predict+students+dropout+and+academic+success>

M.V.Martins, D. Tolledo, J. Machado, L. M.T. Baptista, V.Realinho. (2021) "Early prediction of student's performance in higher education: a case study" Trends and Applications in Information Systems and Technologies, vol.1, in Advances in Intelligent Systems and Computing series. Springer. DOI: 10.1007/978-3-030-72657-7_16

Martins, M. V., Tolledo, D., Machado, J., Baptista, L., & Realinho, V. (2021). Early Prediction of student's Performance in Higher Education: A Case Study. https://doi.org/10.1007/978-3-030-72657-7_16